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Assessing the Robustness of TerraMind to Sensor Domain Gap for On-board Processing on the Φ -sat-2 Mission

Master Thesis

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Abstract

On-board processing promises to overcome the downlink latency and bandwidth constraints that limit the usefulness of Earth observation (EO) data for time-critical applications such as flood and wildfire response. Geospatial Foundation Models (GeoFMs) are attractive candidates for this role because their large-scale, multimodal pre-training yields semantic feature extractors that generalize across tasks with few labeled examples. However, GeoFMs are pre-trained on archived data from missions such as Sentinel-2, and their behavior when exposed to a new sensor is unknown. This thesis investigates whether TerraMind, a Vision Transformer (ViT)-based GeoFM, produces domain-invariant latent representations when applied to real imagery from the European Space Agency’s Φ -sat-2 CubeSat mission, whether a lightweight CNN distilled from it can be made sensor-invariant, and whether doing so is preferable to simply training the compact model on the target sensor directly.

Using a co-registered triplet dataset of real Φ -sat-2, real Sentinel-2, and physics-simulated Φ -sat-2 imagery, we show that the domain gap is present from the encoder’s early layers. This domain gap is measured using embeddings distances of two sensor views of the same ground in TerraMind’s embedding space. Furthermore, a linear probe transfers only a third of its source-domain accuracy to the target sensor. This finding is consistent with a low-level, radiometric origin (differing spectral response functions) rather than a semantic one. Naively fine-tuning on physics-degraded data does not close this gap. Domain-specific batch normalization, applied to the student during cross-architecture knowledge distillation into a lightweight CNN overcomes this gap as it inverts every measurement above (cosine similarity $0.098 \rightarrow 0.948$, probe transfer $33\% \rightarrow 97\%$) while requiring only that the sensor of each input be known, not that inputs be paired across sensors. A paired-contrastive objective exploiting the co-registered pairing directly was also implemented and evaluated, but added no measurable benefit on top of domain-specific normalization alone. Whether this adaptation is actually worth adopting is answered conditionally: the distilled, domain-adapted student outperforms a supervised baseline of identical architecture trained directly on the target sensor at a representative labeled-data budget (mIoU 0.4532 vs. 0.3989), but loses to that same baseline at a much smaller one (mIoU 0.2679 vs. 0.2916), reversing significantly in both directions. A planned few-shot comparison against Φ satNet, an on-orbit GeoFM developed for the same mission, remains to be run.

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Chapter 1

Introduction

Earth observation (EO) satellites now capture more data than ground segments can downlink and process in a timely manner[1]. Ground-based processing is adequate for most applications, but time-critical ones such as flood mapping[2] and wildfire detection[3] require near-real-time inference, since the events they monitor can evolve within hours. Moving the inference step from the ground to the satellite itself removes this constraint. Instead of downlinking large raw imagery for later processing, an on-board model can downlink a small processed data product. This paradigm shift overcomes the downlink bottleneck for time-critical applications, and a growing number of missions have begun demonstrating its technical feasibility[1]. The European Space Agency’s Φ -sat-2 mission is among the most complete demonstrations to date: a 6U CubeSat carrying an 8-band high-resolution multispectral imager alongside a space-qualified Intel Movidius Myriad 2 VPU (Vision Processing Unit)[4]. Rather than running a single fixed application, Φ -sat-2 uses the NanoSat MO Framework[5] to install, update, and orchestrate multiple AI applications on board, much as a smartphone runs apps rather than a single program.

To date, applications running on board Φ -sat-2 have been small, task-specific convolutional neural networks (CNNs), each trained, validated, and uplinked individually for a specific job such as cloud masking, vessel detection, or roadmap extraction[6]. Adding a new capability means repeating the entire pipeline from scratch. Geospatial Foundation Models (GeoFMs) offer a different model of development. Pre-trained on large, diverse archives of EO imagery, a GeoFM such as TerraMind[7] learns a general-purpose semantic representation that downstream tasks can exploit with comparatively few labeled examples, rather than requiring a bespoke architecture and full training run per application. This property matters specifically for applications like flood and burnt-area mapping, where labeled events are rare by nature and a mission cannot simply wait to accumulate a large annotated archive before it becomes useful in an emergency.

Two significant challenges stand between this promise and an operational on-board GeoFM. The first is the architectural constraints due to target hardware limitations. TerraMind is built on the Vision Transformers (ViTs)[8] architecture, which is structurally incompatible with space-qualified accelerators of the Myriad 2’s generation[4], as it predates the hardware and software support that ViT inference requires. The second is the operational domain gap due to sensor differences. GeoFMs such as TerraMind are pre-trained on Sentinel-2 and similar archived missions, but Φ -sat-2 has a physically different sensor, and the statistical properties of its imagery differ from Sentinel-2’s in ways the model never saw during pre-training. Solving either problem alone does not solve the other, as a compressed model built without addressing the domain gap simply compresses

the gap along with it, and a domain-adapted ViT still cannot run on the target hardware.

This thesis addresses both obstacles together, using TerraMind and Φ -sat-2 as the case study, with flood segmentation as the motivating downstream application. The methodological approach combines three elements. First, a physics-based sensor degradation pipeline, based on the simulator developed for the ESA Φ -lab OrbitalAI challenge[9], is used to process Sentinel-2 imagery at the pixel level to simulate the Φ -sat-2 sensor characteristics. This simulation pipeline produces labeled training data for a target sensor that hasn’t accumulated data for flood segmentation yet. Second, a purpose-built dataset of spatially and temporally co-registered image triplets – real Φ -sat-2, real Sentinel-2, and simulated Φ -sat-2, all of the same scene – makes it possible to measure this domain gap directly in latent space and to test candidate adaptation mechanisms against it, rather than relying on downstream accuracy alone. Third, domain-specific batch normalization is applied during knowledge distillation[10] of the TerraMind encoder into a lightweight CNN: the student keeps a separate set of normalization statistics per sensor on otherwise shared weights, so the deployable model adapts to whichever sensor it is actually reading from at inference, since that sensor identity is always known on board. A paired-contrastive alignment objective exploiting the co-registered triplets directly was also implemented and evaluated as an alternative mechanism, but did not improve downstream task performance beyond what domain-specific normalization already provides.

1.1 Problem Definition

The main technical problem is a covariate shift between the sensor domain a GeoFM is pre-trained on and the sensor domain it is tasked to operate on. Let X denote the space of satellite image patches and Y the space of semantic labels (land-cover class, flood/background). Writing p_s and p_t for the source (pre-training, e.g. Sentinel-2) and target (operational, e.g. Φ -sat-2) distributions, a covariate shift is present when

$$p_t(x) \neq p_s(x) \quad \text{while} \quad p_t(y | x) = p_s(y | x). \quad (1.1)$$

In practice, the physical scene and its semantic content are identical across the two domains, but the pixel statistics produced by the sensor observing it differ. This shift occurs because of differing spectral response functions, point spread functions, signal-to-noise characteristics, and geometric band-to-band alignment, all of which are properties of the sensor rather than the scene itself. A model whose learned representation is sensitive to these sensor-specific statistics rather than to the underlying semantics will degrade on the target domain even though nothing about the task itself has changed.

Prior work on GeoFM domain gap has typically measured this shift only indirectly, through downstream task performance, and has compared source and target datasets that differ in geography, acquisition date, and scene content in addition to sensor – confounding the sensor-specific shift this thesis is concerned with, with shifts due to the underlying data distribution itself. The co-registered triplet dataset used in this work is designed specifically to remove that confound, since the real Φ -sat-2, real Sentinel-2, and simulated patches used for evaluation depict the same scene at the same time.

1.2 Research Gap

Two related works describe in similar ways the gap this thesis addresses. Du et al. [11] recently demonstrated an on-orbit GeoFM (a compressed Prithvi variant) on the Kanyini CubeSat and the IMAGIN-e payload, adapting it to the target sensor by retraining task heads on real, though limited, target-domain labels while keeping the encoder frozen. Their measured domain gap, however, is confounded in the way described above: source and target are entirely different benchmark datasets, not co-registered pairs, so sensor-induced shift cannot be isolated from content and geographic shift. Separately, Φ satNet, a bespoke U-Net-based GeoFM developed within the same research group specifically for on-board execution on Φ -sat-2 hardware, sidesteps the ViT-compatibility problem altogether by training a compact, non-transformer architecture from scratch on a purpose-built pre-training corpus, rather than adapting an existing large-scale pre-trained model. Neither work isolates sensor-only covariate shift using co-registered data, and neither tests whether an existing GeoFM’s broad, diverse pre-training transfers to a new sensor with fewer labels than a mission-specific model trained from scratch requires – the specific comparison this thesis undertakes against Φ satNet in Chapter 4.

The gap this thesis addresses is therefore threefold: (i) whether the domain gap between Φ -sat-2 and Sentinel-2-pretrained TerraMind can be measured directly, in latent space, using data that isolates sensor-induced covariate shift from confounding factors; (ii) whether that gap can be closed through an explicit domain-adaptation mechanism rather than incidental exposure to degraded training data, and which such mechanism – a per-domain normalization scheme or a paired-contrastive objective exploiting the co-registered triplets – is actually responsible for closing it; and (iii) whether the resulting invariance survives compression into a hardware-deployable, cross-architecture (ViT-to-CNN) student model, and whether that compression step itself yields a downstream accuracy benefit over training a compact model directly on the target sensor.

1.3 Research Objective and Problem Statement

The objective of this thesis is to determine whether TerraMind can be made to produce domain-invariant latent representations under the Φ -sat-2 sensor domain gap, and whether this invariance can be preserved through knowledge distillation into a lightweight CNN suitable for deployment on the Φ -sat-2 on-board processor, without sacrificing the representational benefits of large-scale GeoFM pre-training. This objective is pursued through three research questions:

1. Does the domain gap caused by sensor differences between Φ -sat-2 and Sentinel-2 measurably affect the latent representations produced by TerraMind, and if so, where in the encoder does this effect concentrate?
2. Can a lightweight CNN distilled from TerraMind be made sensor-invariant given that naive fine-tuning on degraded data alone does not achieve this, and which mechanism is actually responsible if so – domain-specific normalization, a paired-contrastive objective exploiting the co-registered triplets, or both?
3. Does distilling from TerraMind in this way improve downstream Φ -sat-2 task performance over simply training a compact model of the same architecture directly

on the target sensor, so that sensor-invariance and task accuracy are assessed as two separate questions rather than assuming the former implies the latter?

Answering these questions would show that the generalization promise of large GeoFMs – broad semantic transfer from limited labels – is compatible with the low-latency, resource-constrained inference that on-board emergency-response applications require, provided the domain gap is addressed deliberately rather than assumed away.

Chapter 2

Background and Literature Review

This chapter positions the thesis against five parts of the work that each contribute to the overall problem, which is to make the representational benefits of a GeoFM available on board a resource-constrained satellite. Each part is treated in turn, before a closing synthesis states what combination of them remains unaddressed.

2.1 On-board AI for Earth Observation

On-board AI processing has moved from single-purpose CNNs, exemplified by Φ -sat-1’s[12] cloud-detection application[13], toward more general and more capable models, as surveyed by Barretta et al. [1]. ViT architectures have become popular in computer vision due to their ability to capture global context via self-attention mechanisms, often outperforming CNNs on a variety of tasks[14]. This makes ViT-based models strong candidates for on-board EO applications, but their computational and memory requirements are often prohibitive for deployment on resource-constrained satellites[15]. The hardware on board Φ -sat-2 is a constraining factor for the deployment of models, as the CogniSAT-XE1 payload is equipped with the Intel Movidius Myriad 2 VPU [4]. The support for this hardware has been discontinued in 2020, which makes it legacy. The Φ satNet effort reports that the most recent OpenVINO release supporting this hardware is version 2020.3, that experimental testing confirmed only ONNX opset 8–11 are available¹, and that the resulting lack of dynamic-axes support renders transformer-based architectures such as ViT incompatible with the target [16]. These constraints, documented in detail by the Φ satNet effort, are the main motivations for cross-architecture distillation, rather than direct deployment or same-family compression.

2.2 Geospatial Foundation Models

Foundation models are a class of large, pre-trained neural networks that can be adapted to many downstream tasks with minimal additional training[17]. This paradigm has been widely adopted in natural language processing and computer vision, where models such as BERT, GPT, and CLIP have demonstrated remarkable performance across a range of tasks. These models are characterized by their pre-training strategy, which typically involves self-supervised learning on massive, diverse datasets, and by their ability to

¹Supported operations are listed here: https://docs.openvino.ai/2023.3/openvino_docs_ops_opset2.html

generalize across tasks and domains. Once a foundation model is pre-trained, it can be fine-tuned on specific tasks, which makes them highly versatile and efficient for a wide range of applications.

GeoFMs[18] extend the foundation-model paradigm from natural images to Earth observation. Pre-training gives a single large backbone of diverse, multi-modal EO archives so that many downstream tasks can be solved by fine-tuning a small task-specific head rather than training a full model from scratch. TerraMind[7] is a ViT-based, multimodal GeoFM pre-trained on the multimodal TerraMesh dataset[19], which includes Sentinel-2 data. It is explicitly designed to produce transferable semantic representations rather than task-specific features. However, that capacity has, to date, mostly been demonstrated on data drawn from the same sensor family the model was pre-trained on, leaving open how it behaves under different sensor conditions.

2.3 Domain Adaptation

Domain adaptation addresses the general problem of a model trained on a source distribution being applied to a target distribution with different input statistics but unchanged semantics [20]. Within satellite on-board AI specifically, Du et al. [21] demonstrated domain adaptation for satellite-borne cloud detection by retraining task heads on real, though limited, labeled imagery from the Kanyini mission, with the encoder frozen. This line of work was extended on-orbit by Du et al. [11], who deployed a compressed Prithvi-family GeoFM on Kanyini and the IMAGIN-e payload, adapting via real-label head retraining rather than encoder-level invariance training. The domain gap is measured only indirectly through downstream task performance on source and target benchmark datasets that differ in geography and content as well as sensor. This is the central limitation this thesis’s data design addresses, because without co-registered source/target pairs of the same scene, a measured performance drop cannot distinguish sensor-induced covariate shift from confounding differences in what the two datasets depict.

Beyond retraining task heads, approaches to close the gap fall into a small number of families. Their shared theoretical motivation is the bound of Ben-David et al. [22], which limits target-domain error by the sum of the source-domain error, a divergence between the two marginal input distributions, and a term for the error of the best joint hypothesis. Since the first and third terms are not reduced by adaptation, the families below differ mainly in how they attack the divergence term. Discrepancy-based methods penalize a statistical distance between the source and target feature distributions directly, most commonly the maximum mean discrepancy (MMD) [23], which requires only that samples from each domain be available, not that they correspond. Alignment-based methods make the stronger assumption that corresponding inputs are available, and pull their representations together explicitly. X-STARS [24] is the closest known example to the setting of this thesis: it adapts a model pre-trained on one optical sensor to a second sensor through a multi-sensor alignment dense loss, a patch-wise contrastive objective applied between paired acquisitions of the same area by both sensors, within a continual-pretraining framework. It establishes that a paired-contrastive objective is effective when the sensor change is primarily one of spatial resolution, which is not the regime this thesis operates in: the Φ -sat-2/Sentinel-2 gap measured in Chapter 4 is driven mainly by differing spectral response rather than by resolution mismatch.

A separate line of domain-adaptation work sidesteps alignment losses entirely by giv-

ing the model domain-specific normalization statistics instead. Domain-Specific Batch Normalization (DSBN) [25] keeps a single set of convolutional weights but a separate batch-normalization affine transform and running statistics per domain. This only requires that the domain of each example is known, not that examples are paired across domains. This is a weaker requirement than the paired-contrastive or MMD loss objectives discussed above, and, unlike them, is directly applicable wherever the deployed sensor is known at inference.

2.4 Physics-Based Sensor Degradation Simulation

Because labeled data from a satellite sensor does not exist before the satellite is launched, and accumulates slowly afterward for rare events, physics-based simulation of a target sensor’s characteristics from an existing labeled source (typically Sentinel-2) has become a standard tool for pre-mission model development[26][27]. The simulator developed for the ESA Φ -lab OrbitalAI challenge[9] models this transformation as a sequence of physical operations in radiance space – spectral and spatial resampling to the target sensor’s band configuration and ground sampling distance, band-to-band misalignment consistent with push-broom acquisition geometry, signal-to-noise degradation, and optical blur modeled through the sensor’s modulation transfer function – calibrated against real flight data from related missions. This simulator underlies both the degradation pipeline used in this thesis and, independently, the pre-training and benchmarking datasets used by Φ satNet (Section 2.5.1), making it a shared, validated tool within this specific research context rather than a novel contribution of either work.

2.5 Knowledge Distillation Across Architectures

Knowledge distillation (KD)[10] compresses a large "teacher" model into a smaller "student" by training the student to reproduce the teacher’s output distribution. By transferring the teacher’s semantic knowledge (SK), KD allows the student to generalize better on large datasets with a lower computational cost. Some feature-based variants exist that utilize the intermediate representations of the teacher, rather than training the student only against hard labels[28]. T. -D. Le et al. [29] proposed a distillation method between a ViT teacher and a ResNet student for image classification, which uses a dynamic weighting mechanism to adjust each teacher’s influence on the student based on the teacher’s confidence in its predictions. This thesis, on top of cross-architecture distillation – a necessary step given that ViT inference is not supported on the Φ -sat-2 on-board hardware (Section 2.1)– gives the student itself a domain-adaptation mechanism (domain-specific batch normalization, Section 3.5). This mechanism ensures that the student can learn domain-specific characteristics rather than relying on passive inheritance of the invariance the teacher might have than been able to learn. A paired-contrastive alignment objective exploiting the co-registered triplets was implemented and evaluated as an additional mechanism on top of this, but was found not to improve downstream task performance beyond what domain-specific normalization already provides (Section 3.4).

2.5.1 Φ satNet

Φ satNet is a U-Net-derived, multi-task-pretrained GeoFM being developed by the Φ -lab specifically for on-board execution on Φ -sat-2, using a purpose-built pre-training corpus and benchmark suite built on the same OrbitalAI simulator described above. By training a compact, non-transformer architecture from scratch, Φ satNet avoids the ViT-hardware-incompatibility problem entirely, and its authors report that it matches or exceeds unconstrained transformer-based GeoFMs on several on-board tasks [16]. This is a different design philosophy from the one pursued here, not a competing solution to the same problem. Φ satNet optimizes for a hardware-native architecture trained on a curated, mission-specific task suite, while this thesis asks whether an existing, diversely pre-trained GeoFM can be adapted to new tasks with fewer labels than training a mission-specific model requires.

One property of Φ satNet bears directly on the problem this thesis addresses. Its pre-training method and its n -shot benchmarks are built entirely on simulated Φ -sat-2 imagery derived from Sentinel-2, not on real acquisitions [16]. Φ satNet therefore does not escape the sensor domain gap by being mission-specific; it relocates it. Its reported few-shot advantage rests on the assumption that physically simulated imagery stands in adequately for the real sensor, and its own evidence base, being simulated throughout, cannot test that assumption. The sim-to-real question is thus one that the bespoke and the adapted approach inherit alike, and co-registered real acquisitions are what either would need in order to answer it. A matched few-shot comparison against Φ satNet on a shared task and label budget remains the appropriate test of the label-efficiency claim; Chapter 4 returns to this positioning, but that comparison has not been run and is left to future work.

2.6 Synthesis

No work cited above isolates sensor-induced covariate shift using co-registered source/target data, gives a cross-architecture distillation student an explicit, domain-aware adaptation mechanism rather than relying on incidental exposure to degraded or limited real data, and verifies that this invariance – not merely task accuracy – survives compression into hardware-deployable form. The spatially and temporally co-registered Φ -sat-2/Sentinel-2/simulated triplet dataset used in this thesis is what makes the first of these possible; the domain-specific batch normalization applied during cross-architecture distillation (Chapter 3) is what makes the second and third possible, with the co-registered triplets additionally serving as the diagnostic instrument that shows a paired-contrastive alternative does not do better. That real-acquisition component also separates this work from the mission-specific alternative: Φ satNet is pre-trained and benchmarked on simulated Φ -sat-2 imagery, so the assumption that simulation stands in for the real sensor is one both approaches make and only real, co-registered data can check.

Chapter 3

Methodology

This chapter presents the complete analytical workflow: the data used and the role each dataset plays, the physics-based degradation pipeline and its calibration, the protocol for measuring the domain gap, the domain-adaptation mechanism adopted (and the one rejected, with the evidence for rejecting it), the cross-architecture distillation procedure, and the evaluation protocol. It closes with a dedicated account of the tooling and reproducibility practices used throughout, as required by the thesis guidelines.

The data underlying this work is split into two categories. The first is the co-registered triplet dataset itself: real Φ -sat-2 acquisitions, obtained via the ESA Insula Perception platform, matched to concurrent Sentinel-2 scenes and to a physics-simulated Φ -sat-2 rendering of the same Sentinel-2 scene, co-registered using the LightGlue deep feature-matching algorithm [30] and filtered for spatio-temporal proximity and cloud cover. This dataset carries land-cover labels derived from ESA WorldCover 2021, providing the only currently available real-sensor ground truth. The second category is simulated: a physics-degraded, graded version of the Sen1Floods flood-segmentation benchmark, produced with the same degradation pipeline, standing in for a real labeled Φ -sat-2 flood dataset that does not yet exist because the mission has not captured an annotated flood event.

The overall workflow followed a prototyping-implementation-refinement logic rather than a fixed plan executed once. An initial conceptual model – fine-tune TerraMind on real Φ -sat-2 labels, measure the gap, close it – was implemented, evaluated against intermediate diagnostics (embedding visualizations, similarity metrics), and revised twice on the basis of what those diagnostics showed: first when a normalization defect was found to be corrupting an early performance baseline (Section 3.3.1), and second when a first domain-adaptation mechanism was found, empirically, not to work (Section 3.4.1), motivating the mechanism actually adopted (Section 3.4.3).

3.1 Data

3.1.1 The co-registered triplet dataset

The main dataset is a collection of spatially and temporally co-registered image triplets, produced in collaboration with the ESA Φ -lab: real Φ -sat-2 acquisitions (obtained via the ESA Insula Perception platform), the corresponding real Sentinel-2B scene, and a physics-simulated Φ -sat-2 rendering of that same Sentinel-2 scene (Section 3.2). Candidate scenes were filtered by spatio-temporal proximity between the Φ -sat-2 and Sentinel-2 overpasses and by a cloud-cover threshold on both acquisitions, and precise pixel-level alignment

was achieved using the LightGlue deep feature-matching algorithm [30], which identifies corresponding semantic keypoints across the two sensors despite their differing spatial and radiometric characteristics.

This dataset carries labels for a single task: land-use/land-cover (LULC) classification, derived from ESA WorldCover 2021 and rasterized onto each triplet’s real- Φ -sat-2 footprint. Its role in this thesis is twofold: it is the only real-sensor ground truth available for any downstream task, and its co-registration is what makes it possible to measure sensor-induced covariate shift in TerraMind’s latent space (Section 3.3) without the scene-content and geography confounds present in prior domain-gap studies that compare unrelated source and target datasets.

Figure 3.1 shows four such triplets. Qualitatively, the real acquisitions are consistently grainier than their simulated or source counterparts, and occasionally differ subtly in overall tone; the simulated column is, by construction, closer in appearance to Sentinel-2 than to the real sensor it is meant to approximate, a point returned to below.

Scale and coverage

After discarding a small number of corrupted source products, the dataset comprises 253,228 co-registered triplet patches drawn from 1,299 distinct Φ -sat-2 products (individual acquisitions), a median of 209 patches per product (up to 713 for the largest). Acquisitions span 2024-08-21 to 2026-04-15 (21 months), with collection volume markedly higher from mid-2025 onward than in the initial ramp-up period. Real- Φ -sat-2 and Sentinel-2 acquisition dates are matched to within a median of 6.8 days and a maximum of 14.9 days, not simultaneously, which is an acceptable temporal offset for a largely stable target such as land cover but a caveat worth stating rather than assuming away. 73.1% of retained patches have under 5% total cloud cover, and only 0.199% of pixels are unlabelled after the ESA WorldCover rasterization step.

Geographically, the 1,299 source products span latitudes -54° to 76° with global but uneven coverage, concentrated over western North America, the southern Andes, the Mediterranean basin and Middle East, and parts of East and South Asia (Figure 3.3). By Köppen–Geiger climate group, the patch population is dominated by arid (41%) and temperate (29%) climates, with tropical (14%), continental (10%), and polar (1%) regions comparatively under-represented (4% unclassified).

The eleven ESA WorldCover classes present are similarly imbalanced by construction rather than by sampling choice: grassland (20.8%), tree cover (20.0%), bare/sparse vegetation (16.2%), built-up (14.4%), and cropland (13.4%) account for the large majority of labelled pixels, while snow/ice, mangroves, and moss/lichen each fall below 0.6% (Figure 3.4). The typical patch is fairly homogeneous – a single class occupies a median of 73% of any given patch’s labelled area.

Train/validation/test split and a limitation

The dataset is split 80/10/10 (202,582 / 25,323 / 25,323 patches) at the *patch* level with a fixed seed, and the resulting class-prior distribution is stable across all three splits, confirming the split does not introduce label-distribution shift (Figure 3.5, left). It does, however, introduce a different problem: because splitting is patch-level rather than product-level, all 1,299 source products are represented in the training set, and every one of the 1,287 distinct products contributing to the test set also contributes patches to training (Figure 3.5, right). Since patches from the same product are spatially adjacent

Co-registered views, independently contrast-stretched

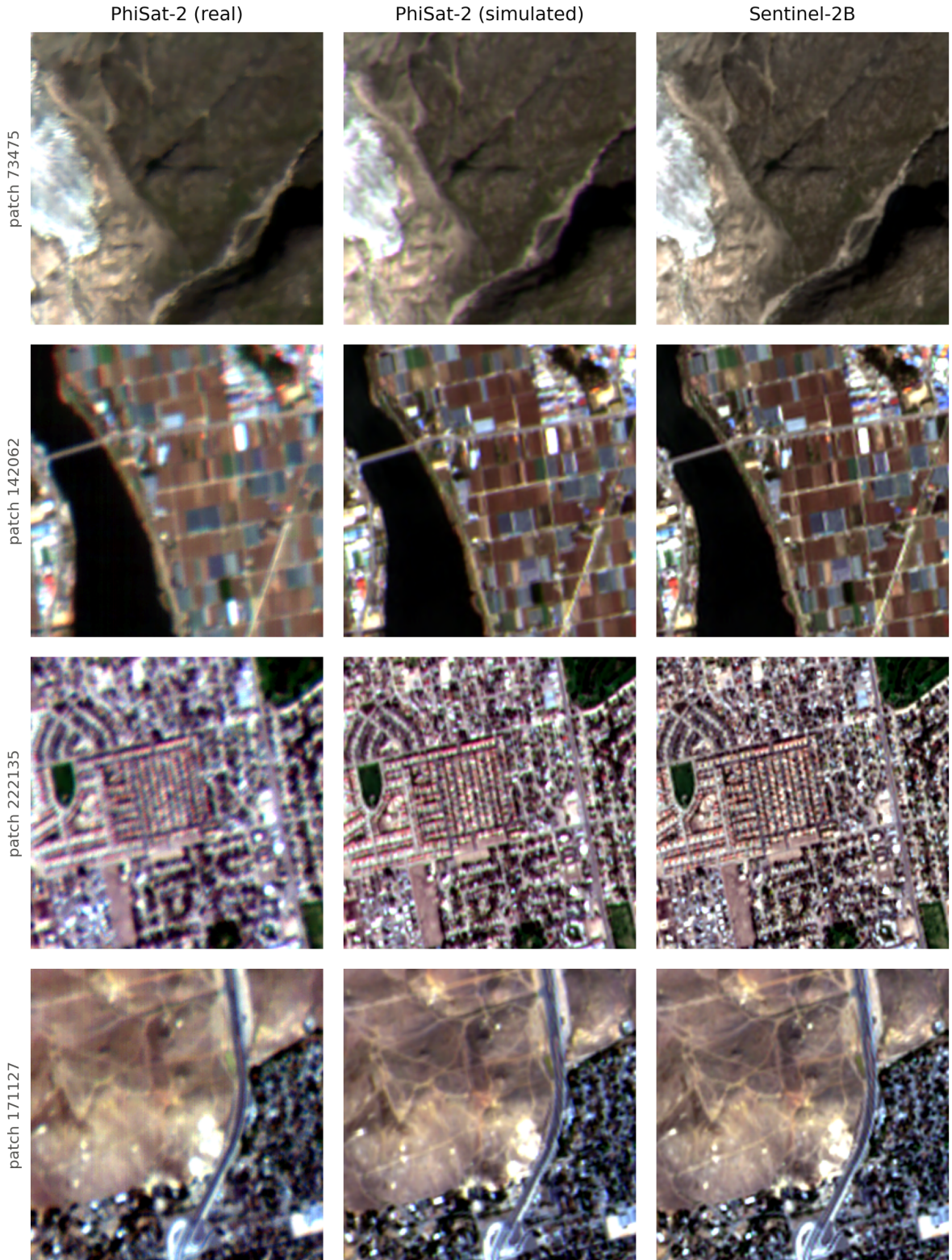


Figure 3.1: Four representative triplets, each row showing the same scene as observed by real Φ -sat-2, the physics-simulated Φ -sat-2 rendering, and Sentinel-2B (each channel independently contrast-stretched for display). The real acquisitions show visibly grainier, noisier texture than either the simulated or source Sentinel-2 view – most apparent in the dark water/shadow region of patch 171127 – consistent with the sensor noise characteristics the degradation pipeline (Section 3.2) is designed to reproduce.

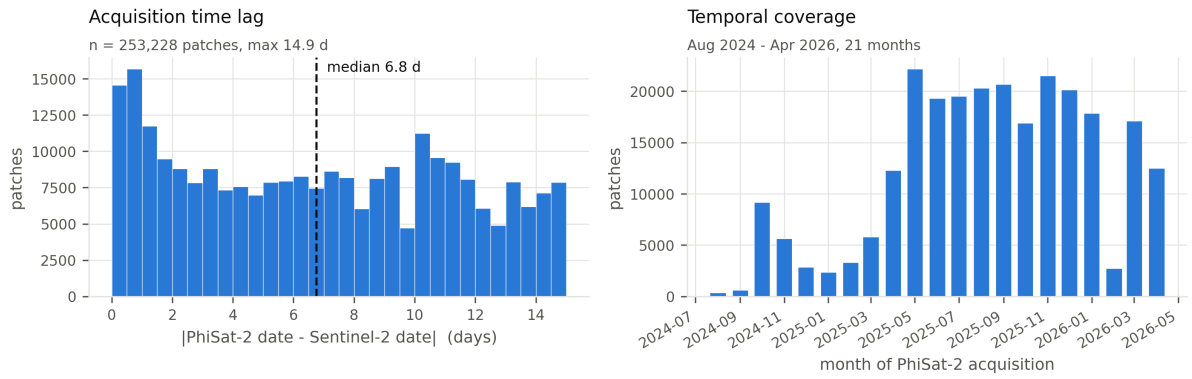


Figure 3.2: Left: distribution of the absolute time lag between matched Φ -sat-2 and Sentinel-2 acquisitions (median 6.8 days, maximum 14.9 days, $n = 253,228$). Right: monthly acquisition volume, Aug. 2024 – Apr. 2026, showing the collection ramp-up in the first eight months.

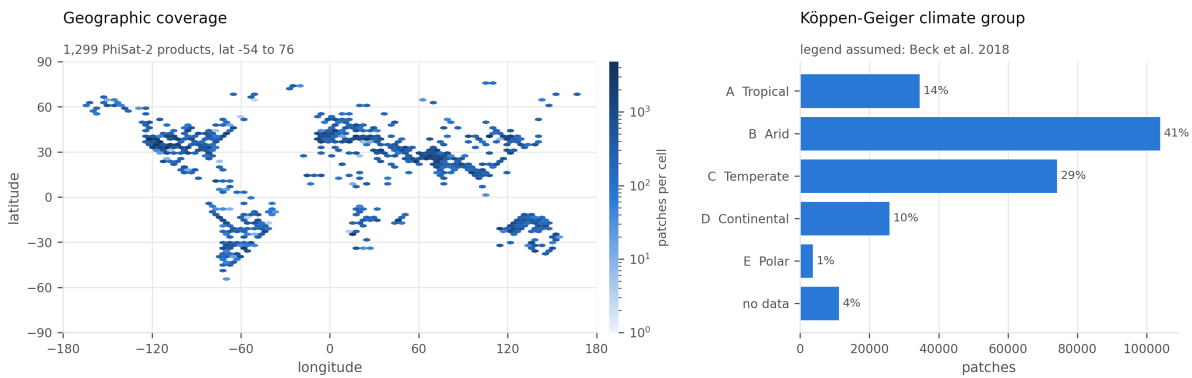


Figure 3.3: Left: spatial density of Φ -sat-2 products (hexbin, log-scaled patch count per cell). Right: distribution of patches by Köppen–Geiger climate group (Beck et al., 2018 classification).

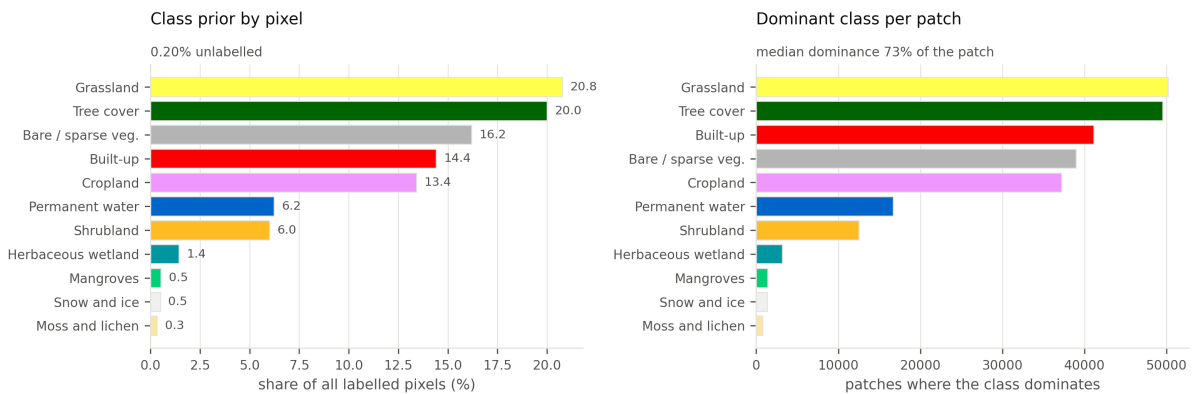


Figure 3.4: Left: share of all labelled pixels by ESA WorldCover class. Right: number of patches in which each class is the dominant (plurality) class.

or overlapping crops of one acquisition, this means the test set is not a clean geographic or scene-level hold-out, and real-data performance figures reported in Chapter 4 should be read as an optimistic upper bound on generalization to genuinely unseen scenes rather than as a strict out-of-sample estimate. A product-level (rather than patch-level) re-split is the direct remedy and is noted as future work in Section 4.6.

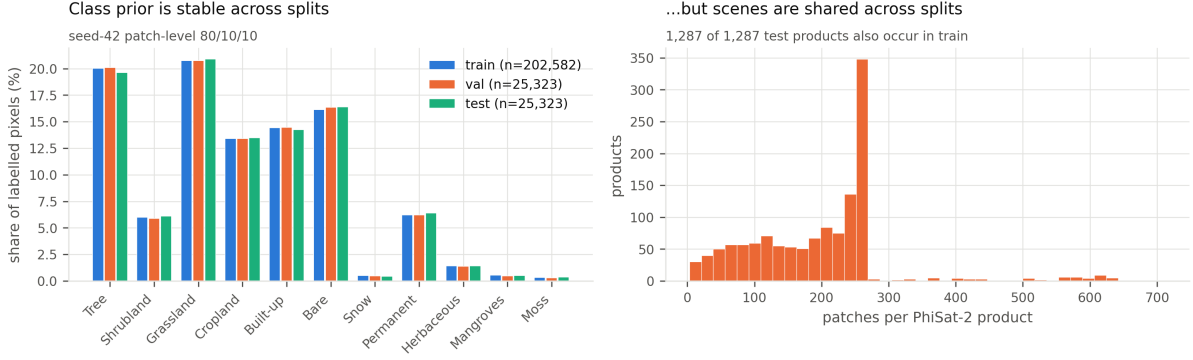


Figure 3.5: Left: class prior is stable across the train/validation/test split. Right: every one of the 1,287 products contributing to the test set also contributes patches to the training set – the split is patch-level, not scene-level.

Measured sensor discrepancy

Three independent measurements quantify the sensor discrepancy this dataset is designed to expose, prior to any model being involved.

First, the raw radiometric scales of the two sensors differ by roughly an order of magnitude (Figure 3.6): real Φ -sat-2 digital numbers have per-band medians of roughly 55–150 (12-bit range, with measurable saturation of 2.1–2.2% of pixels in the RE1/RE2 bands and dropout of 0.3–0.36% in RE2/RE3), against Sentinel-2 and simulated- Φ -sat-2 medians around 2,200–3,300 (consistent with the L1C TOA-reflectance-scaled DN convention). The simulated and raw Sentinel-2 domains are close to statistically indistinguishable at this level (their boxes are visually superimposed in Figure 3.6), confirming that the physics-based degradation pipeline (Section 3.2) perturbs fine spatial and spectral structure rather than gross intensity statistics – the domain gap this thesis is concerned with is a real-vs-simulated gap, not a simulated-vs-source gap.

Per-domain normalization (Section 3.3.1) closes most, but not all, of this gap (Figure 3.7): after z -scoring each sensor independently, the three distributions are centered similarly, but real Φ -sat-2 retains visibly wider spread and longer tails in every band than Sentinel-2 or the simulated domain. This residual spread difference, not the location/scale difference normalization already removes, is what the domain-adaptation mechanism of Section 3.4 must close. The statistics underlying this figure were measured on the sampled patches shown here, which differ slightly from the constants currently hardcoded in the dataset loader; the loader’s normalization constants should be refreshed from the full measured statistics before the next training run, so that all reported results use the same normalization the figures were generated from.

Second, per-band correlation between co-registered real- Φ -sat-2 and Sentinel-2 pixels is only weak-to-moderate even for matched same-scene observations (Pearson r 0.31–0.54, Spearman ρ 0.37–0.61 across the seven shared bands; Figure 3.8), with the weakest

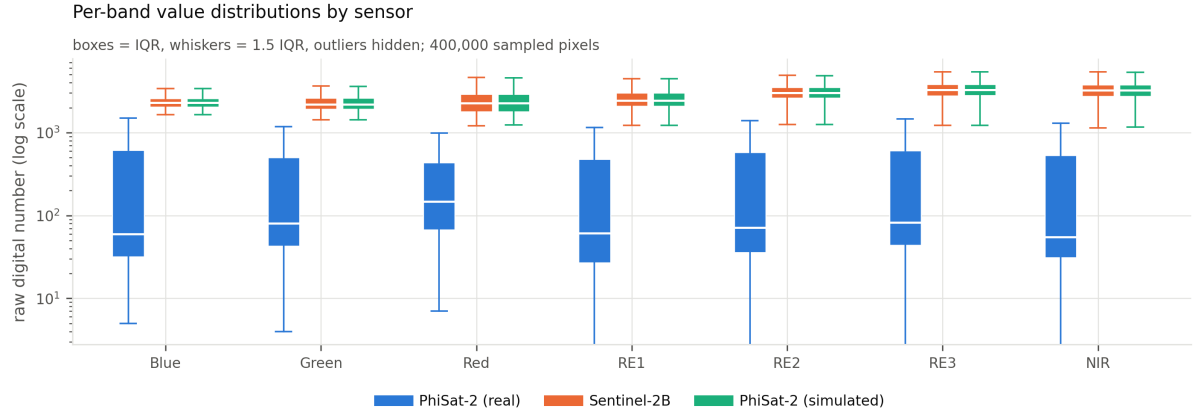


Figure 3.6: Per-band raw digital number distributions by sensor (log scale; boxes = IQR, whiskers = 1.5×IQR, outliers hidden; 400,000 sampled pixels). Real Φ -sat-2 occupies a scale roughly one order of magnitude below Sentinel-2 and its simulated counterpart, which closely overlap each other.

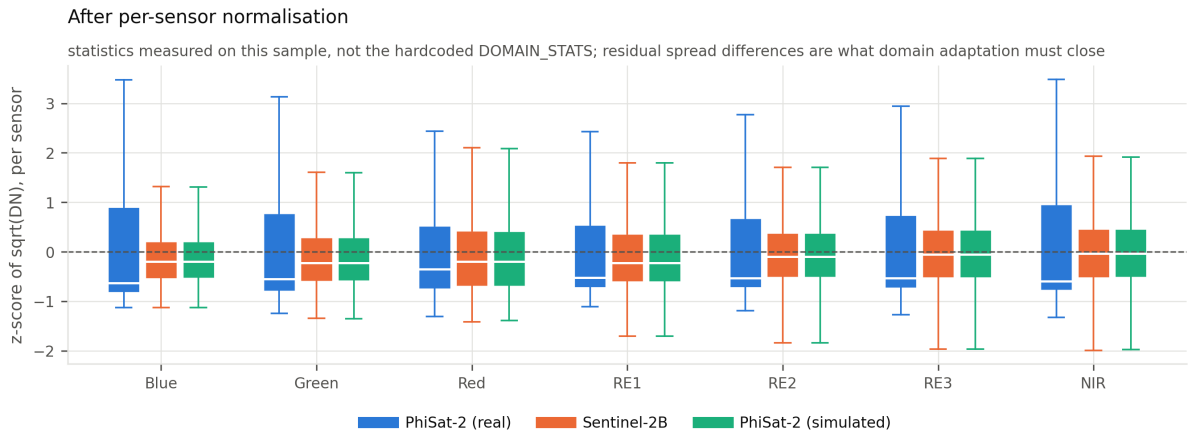


Figure 3.7: The same per-band distributions after per-sensor z -score normalization of $\sqrt{\text{DN}}$. Location and scale are aligned, but real Φ -sat-2 retains a wider spread than Sentinel-2 or the simulated domain in every band – the residual gap normalization alone cannot close.

correspondence in the red-edge bands (RE1, RE2) and the strongest in the red band. This is a direct, model-free demonstration that the two sensors disagree substantially about the same physical target observed within the same fortnight. For contrast, the equivalent cross-band correlation between the simulated domain and its Sentinel-2 source is near-perfect (Spearman ρ 0.98–1.00 on matching bands; Figure ??), as expected since the simulated patches are deterministic transformations of the same Sentinel-2 pixels – the gap between real and simulated Φ -sat-2 is therefore not explained by any weakness in the correlation analysis itself.

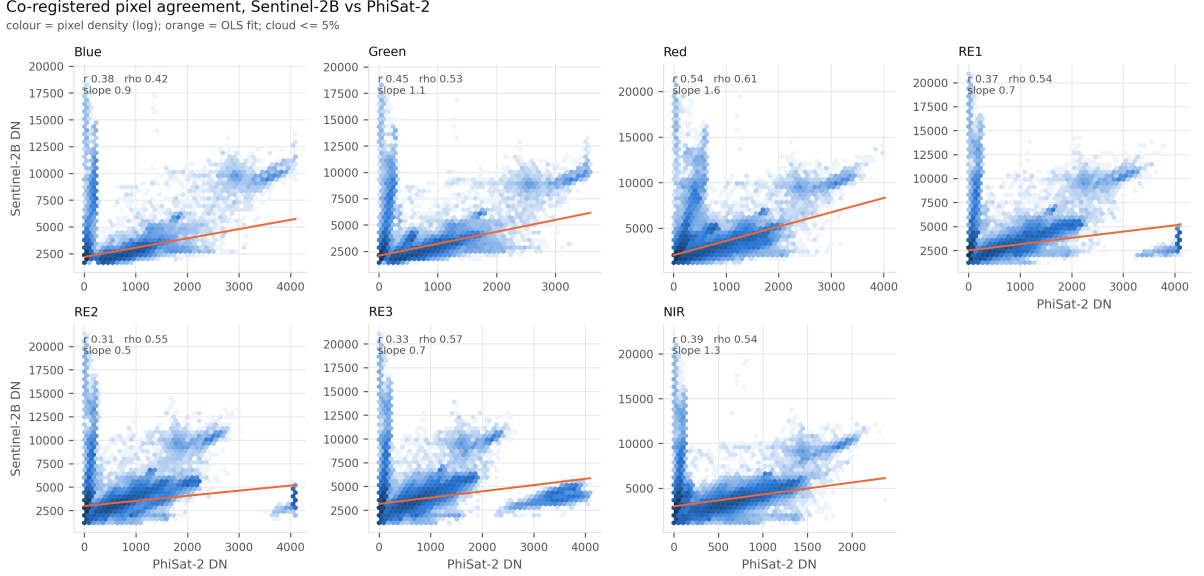


Figure 3.8: Per-band pixel agreement between co-registered real- Φ -sat-2 and Sentinel-2B digital numbers (hexbin density, log scale; orange = ordinary least-squares fit; patches with $\leq 5\%$ cloud only). Point clouds are diffuse around the fitted line in every band, and several bands show a distinct vertical streak near zero Φ -sat-2 DN spanning the full Sentinel-2 DN range, consistent with localized real-sensor dropout or saturation artifacts rather than measurement noise alone.

Third, and most directly relevant to the degradation-driver hypothesis discussed in Chapter 4, the *shape* of the spectral signature – each sensor’s per-band median normalized by its own overall band mean, so that absolute scale is divided out – differs markedly between real Φ -sat-2 and Sentinel-2/simulated- Φ -sat-2 (Figure 3.9). Sentinel-2 and the simulated domain are, again, visually indistinguishable from each other. Real Φ -sat-2, in contrast, shows a pronounced peak at the red band (1.85 relative to its own mean, against 0.85 for Sentinel-2/simulated) and a dip at the near-infrared band (0.70 against 1.21), a difference in relative spectral shape, not merely absolute intensity. This is direct, quantitative support for attributing the domain gap primarily to spectral response function mismatch (Section 4.2) rather than to spatial resolution or noise level, since a resolution- or noise-driven gap would not be expected to alter the relative shape of the spectral signature this distinctly.

These measured per-domain statistics (mean, standard deviation, and clip value per band, computed separately for the real, Sentinel-2, and simulated domains) are what parametrize the per-domain normalization protocol of Section 3.3.1.

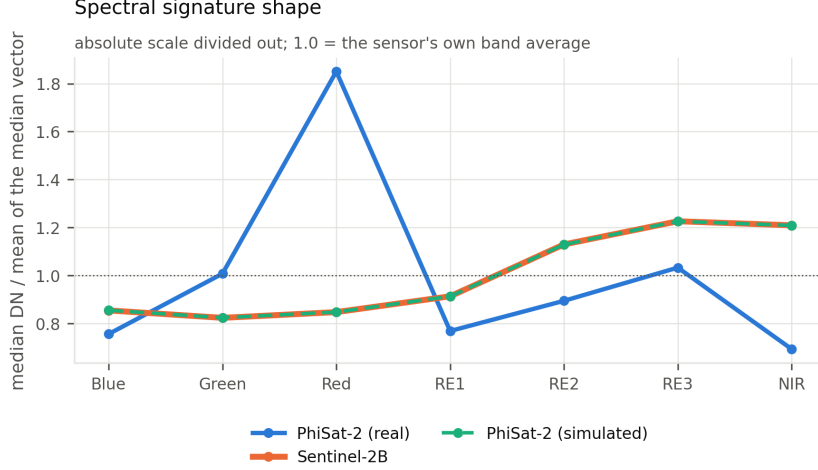


Figure 3.9: Median per-band digital number, normalized by each sensor’s own overall band mean (1.0 = that sensor’s average). Sentinel-2B and simulated Φ -sat-2 overlap almost exactly; real Φ -sat-2 has a markedly different spectral shape, peaking at Red and dipping at RE1 and NIR.

3.1.2 Simulated degraded flood-segmentation dataset

No real Φ -sat-2 flood dataset exists, because the mission has not yet captured and labeled a flood event; this is a structural limitation of any new mission, not a data-collection oversight. To obtain labeled data for the flood-segmentation task motivating this thesis, the Sen1Floods benchmark is passed through the same degradation pipeline used to build the triplets, producing a graded ladder of Φ -sat-2-like flood scenes at increasing degradation severity. This simulated dataset plays a different role from the triplets: it is the training and evaluation substrate for the flood task itself, not the substrate for measuring the domain gap, since it lacks a real- Φ -sat-2 counterpart against which to validate the simulation directly.

3.2 Physics-Based Sensor Degradation Pipeline

The degradation pipeline adapts the simulator developed for the ESA Φ -lab OrbitalAI challenge [9], which projects labeled Sentinel-2 L1C tiles into the Φ -sat-2 sensor domain by modeling the physical differences between the two instruments as a sequence of operations in radiance space. Because these effects are defined physically on radiance, not reflectance, the pipeline converts source reflectance to top-of-atmosphere radiance, applies each degradation in turn, and converts back to reflectance to yield a simulated L1C product. Pixel coordinates are written (i, j) and spectral band index b_n throughout.

Radiance–reflectance conversion. TOA reflectance ρ and TOA radiance L are related by

$$\rho(i, j, b_n) = \frac{\pi d^2}{E_{\text{SUN}}(b_n) \cos \theta_S(i, j)} L(i, j, b_n), \quad (3.1)$$

where d is the Earth–Sun distance in astronomical units, $E_{\text{SUN}}(b_n)$ the exo-atmospheric solar irradiance for band b_n , and $\theta_S(i, j)$ the solar zenith angle at pixel (i, j) . The

pipeline enters via the inverse of Equation (3.1) and exits by re-applying it after all radiance-domain degradations; solar irradiance and zenith angle are read from Sentinel-2 tile metadata via the Copernicus Data Space Ecosystem OData API, and no additional solar-irradiance adjustment is needed because Sentinel-2 and Φ -sat-2 spectral bands are close enough in wavelength. Earth–Sun distance is computed from acquisition day-of-year as $d = 1 - 0.01672 \cos(0.9856(day - 4) \cdot \pi/180)$.

Spatial resampling. Φ -sat-2 has a uniform ground sampling distance of 4.75 m across all bands; Sentinel-2 provides 10 m (bands B2, B3, B4, B8) and 20 m (bands B5, B6, B7). Each band is resampled to the common 4.75 m grid using bicubic interpolation, which limits the resampling artifacts that bilinear interpolation would introduce, though it cannot recover spatial detail the coarser source bands never captured.

Band-to-band misalignment. Push-broom acquisition co-registers spectral bands imperfectly, since detector rows are exposed at slightly different times and platform micro-vibration perturbs the line of sight during each band’s integration window. For the raw L1A product, the displacement of band b_n relative to the previously acquired band follows

$$\|\Delta \mathbf{r}(b_n)\| \sim \mathcal{N}(\mu_{b_n}, \sigma^2), \quad \mu_{b_n} \in [0.44, 1.1] \text{ pixels}, \quad (3.2)$$

with direction sampled uniformly over $[0, 2\pi)$ and displacement accumulating across preceding bands. For the geometrically corrected L1C product used throughout this thesis, an on-board co-registration step removes this accumulation, and the residual is instead modeled as a single random displacement of each band relative to a reference band ($\mu_{b_n} = 0$), with standard deviation set higher over water ($\sigma_{\text{sea}} = 6$ pixels) than over land ($\sigma_{\text{land}} = 1$ pixel), reflecting the harder auto-registration problem posed by feature-poor sea surfaces.

Signal-to-noise degradation. The baseline formulation draws a per-band noise sample from $\mathcal{N}(0, 1)$ scaled by the ratio of a reference radiance to a specified SNR,

$$L_{\text{noise}}(i, j, b_n) = \frac{L_{\text{ref}}(b_n)}{\text{SNR}(b_n)} \mathcal{N}(0, 1), \quad (3.3)$$

which implicitly treats noise variance as signal-independent, matching the specified SNR only near L_{ref} . A signal-dependent variant, consistent with a shot-noise regime in which SNR scales with the square root of collected signal, rescales the reference SNR to an effective, radiance-dependent value and samples

$$L_{\text{noise}}(i, j, b_n) = \frac{\sqrt{L_{\text{in}}(i, j, b_n) L_{\text{ref}}(b_n)}}{\text{SNR}(b_n)} \mathcal{N}(0, 1). \quad (3.4)$$

The implementation used for the calibrated baseline degradation level applies Equation (3.4). For the extended sweep used to push beyond nominal Φ -sat-2 noise conditions – needed to stress-test domain-invariance under more severe degradation than the flight-validated parameters cover – a simplified variant samples a base SNR uniformly from a specified range and scales noise by the ratio of that SNR to the batch-wise maximum radiance, trading exact photometric fidelity for the ability to sweep degradation severity beyond the calibrated regime. This trade-off is stated explicitly here because it is a deliberate design choice, not an approximation error: the baseline level is flight-parameter-calibrated, and only the extended sweep sacrifices fidelity for range.

Optical blur via MTF/PSF. Optical blur is modeled through the system modulation transfer function (MTF), aggregating static (optics, detector), dynamic (jitter, platform instability), and smearing (motion) contributions. Assuming shift-invariant blur, imaging reduces to convolution with the system point spread function (PSF). Given the MTF amplitude at Nyquist frequency $M(b_n)$, the MTF is modeled as Gaussian in normalized spatial frequency k ,

$$\text{MTF}(b_n, k) = e^{\ln M(b_n) \cdot k^2}, \quad (3.5)$$

with the corresponding PSF obtained as its inverse Fourier transform and convolved with the radiance image. For Φ -sat-2 the average Nyquist MTF is $M_\Phi(b_n) \approx 0.05$; for Sentinel-2, $M_{S-2}(b_n) \approx 0.30$. The reference simulation applies Equation (3.5) directly with $M_\Phi(b_n)$, leaving the intrinsic Sentinel-2 blur uncorrected; a combined-filter form exists for the case where source blur should also be compensated, but is not required here since the intrinsic Sentinel-2 blur is small relative to Φ -sat-2’s.

This pipeline is shared infrastructure within the research group: it also underlies the pre-training and benchmark datasets used by Φ satNet (Section 2.5.1). The novelty of this thesis lies in how the pipeline’s output is used – to isolate and then close sensor-specific covariate shift via co-registered pairs – not in the pipeline itself.

3.3 Domain Gap Measurement

3.3.1 Normalization protocol

Raw Φ -sat-2 digital numbers are heavy-tailed; a square-root transform is applied first to bring the distribution closer to Gaussian, followed by a smooth (soft-saturating) clip toward a domain-specific clip value rather than a hard clip, and finally per-domain z -score standardization using mean and standard deviation statistics computed separately for each sensor domain (real Φ -sat-2, real Sentinel-2, simulated Φ -sat-2), rather than a single shared normalization applied across domains. An early baseline was corrupted by a normalization defect that caused weight saturation and an artificially depressed performance estimate; correcting it was a precondition for every result reported in Chapter 4.

Because per-domain normalization could, in principle, itself account for an apparent domain gap that is really just a difference in raw value scale, the domain-gap measurement below was run both with and without per-domain normalization. The two conditions produced similar overall results, and per-domain normalization in fact produced *lower* cosine similarity between domains in the encoder’s early layers – the opposite of what a normalization artifact would predict – indicating that the measured gap in early layers is a genuine representational effect rather than a calibration artifact.

3.3.2 Direct and indirect evidence

The domain gap is measured directly, in latent space, using the co-registered triplets: paired cosine similarity between the embeddings of the two sensor views of the same ground, proxy $\mathcal{A}$ -distance (PAD) computed per encoder layer to locate where the gap concentrates, and a cross-domain linear-probe-transfer test in which a probe fitted on one sensor’s embeddings is applied to the other’s embeddings of the same patches. It is measured indirectly through downstream task performance: LULC mIoU of the distilled student relative to a supervised baseline of identical architecture trained directly on the

target sensor (Section 3.6), and flood-segmentation IoU across the simulated degradation ladder of Section 3.1.2.

3.4 Domain-Adaptation Mechanism

3.4.1 A rejected mechanism: naive exposure to degraded data

The first mechanism attempted was implicit: fine-tune TerraMind’s encoder on the LULC task using degraded, simulated training data, on the premise that exposure to degraded examples during training would itself induce a more domain-invariant representation. Cross-validation against the real- Φ -sat-2/Sentinel-2 triplets showed this does not hold: a model fine-tuned this way shows no lower domain gap to real Φ -sat-2 than a model fine-tuned only on clean Sentinel-2. Standard supervised fine-tuning, even on physics-degraded data produced by a flight-calibrated simulator, applies no explicit pressure toward a sensor-invariant representation, and none emerges as a side effect. An MMD-minimization loss applied directly to the encoder during this fine-tuning stage was also explored as a more explicit alternative and is not pursued further in this thesis, superseded by the mechanism below, which exploits a structural asset – co-registration – that a generic MMD objective does not use.

3.4.2 Adopted mechanism: domain-specific batch normalization

The mechanism ultimately adopted requires a much weaker assumption than either of the above: only that the domain of each training example is known, not that examples be paired across domains. Domain-Specific Batch Normalization (DSBN) [25] gives the student a separate batch-normalization affine transform and running statistics per sensor domain, on top of otherwise fully shared convolutional weights. Concretely, the two-branch distillation scheme (Section 3.5) forwards a Sentinel-2 batch and a Φ -sat-2 batch through the student separately in each training step; DSBN lets each pass use its own domain’s normalization statistics rather than a single statistics set blended across both. Because the deployed sensor on board Φ -sat-2 is always known at inference, selecting the matching normalization branch at deployment time costs nothing operationally, unlike a sensor-agnostic representation, which would need to be invariant without ever being told which sensor produced a given input.

3.4.3 Ablated mechanism: paired-contrastive triplet alignment

The co-registered triplet dataset provides something most domain-adaptation settings lack: for every training example, the identical underlying scene observed by two or three different sensors. This licenses a directly supervised alignment objective rather than an unpaired one: a contrastive loss that pulls the embeddings of matched real- Φ -sat-2, real-Sentinel-2, and simulated- Φ -sat-2 patches of the same scene together, while pushing apart embeddings of unrelated scenes, regardless of which sensor produced each embedding. This directly penalizes the encoder for representing sensor identity, rather than only scene content, and does so using the strongest available supervisory signal – exact scene correspondence – instead of matching marginal distributions between unpaired sensor domains as an MMD-style objective would.

This objective was implemented and evaluated on top of the DSBN configuration above, with the co-registered triplets used exactly as designed: to test a training mechanism, not only to measure the gap. On the LULC task, it did not improve downstream performance beyond the DSBN-only configuration (Section 4.3), and all headline results in Chapter 4 are reported with its weight set to zero. This is reported as a negative ablation result rather than omitted: the co-registered pairing that makes a contrastive objective possible is a genuine structural asset of this dataset, and the fact that exploiting it directly added nothing measurable on top of a much simpler per-domain normalization scheme is itself informative about where the sensor gap actually lives (Section 4.2) and about how much machinery is actually needed to close it.

3.5 Cross-Architecture Knowledge Distillation

TerraMind’s ViT-based encoder cannot run on the Φ -sat-2 on-board processor: the Myriad 2 VPU’s supported operator set has no dynamic-axes support, which ViT inference requires regardless of model size. Compression must therefore cross architecture families, from ViT teacher to CNN student, rather than compress within the ViT family as some prior on-board GeoFM work does. The teacher is TerraMind, fine-tuned on the Sentinel-2 branch of the triplet dataset for the LULC task and then frozen; the student is a U-Net[31] trained from random initialisation.

Two placements for a domain-adaptation mechanism were considered: baking invariance into the frozen teacher and letting the student inherit it passively by mimicking the teacher’s outputs, or giving the student its own adaptation mechanism directly. This thesis adopts the second, for a reason independent of which specific mechanism is used: the student, not the teacher, is what is actually deployed, so whatever mechanism is used should be evaluated on the student’s own representation rather than assumed to transfer from the teacher. Two candidate mechanisms were implemented at the student level: domain-specific batch normalization (Section 3.4.2), adopted as the mechanism responsible for every headline result in Chapter 4, and a paired-contrastive triplet loss (Section 3.4.3), evaluated as an ablation on top of it and found not to improve downstream performance further. DSBN also incidentally repairs a train/test inconsistency inherent to the two-branch training scheme: the two sensors are forwarded in separate passes, so during training each pass would otherwise be normalized by its own domain’s batch statistics while a single shared set of running averages – a blend of both domains – is used at inference, meaning the encoder would never encounter, at test time, the normalization under which it was optimised. Giving each domain its own running statistics closes this gap as a side effect of the same mechanism.

The distilled student is therefore domain-*adaptive* rather than domain-*invariant* in the strict sense: convolutional weights are shared across sensors, but normalization statistics are not, and the correct branch must be selected at inference according to which sensor produced the input. This is a weaker property than a single sensor-agnostic representation, but it is the property an on-board deployment actually needs, since the acquiring instrument is always known at inference time – unlike, for instance, a model expected to generalize to sensors not seen during training at all.

3.6 Evaluation Protocol

Four evaluations follow from the mechanisms above, corresponding to the three research questions of Section 1.3. First (RQ1), whether the sensor gap is visible in the teacher’s representation itself: paired cosine similarity, per-layer PAD, and cross-domain linear-probe transfer between the two sensor views of the same co-registered patches. Second (RQ2), whether the distilled student’s representation inverts these measurements relative to the teacher, and which of the two candidate mechanisms (DSBN, contrastive alignment) is responsible. Third (RQ3), whether distillation with the adopted mechanism improves LULC task performance over a supervised baseline of identical student architecture trained directly on labelled Φ -sat-2 with no teacher and no second sensor, at a label budget large enough to be representative and, separately, at a much smaller budget, since the direction of this comparison is not assumed to be constant across label budgets. Fourth, flood segmentation across the simulated degradation ladder, to characterize how gracefully performance degrades with increasing sensor-noise severity. A few-shot label-efficiency comparison against Φ satNet (Section 2.5.1) on matched small label budgets, testing directly whether adapting TerraMind’s broad, diverse pre-training generalizes to Φ -sat-2 with fewer labels than Φ satNet’s mission-specific pre-training requires, remains planned but has not been run (Section ??). The earlier protocol of fully fine-tuning TerraMind directly on real Φ -sat-2 LULC labels, evaluated on the same real domain without any cross-sensor test, produced a baseline mIoU that is retained in Appendix ?? for reference but is superseded by the source-vs-target evaluation above, which is what the research questions actually require.

3.7 Tools, Software Environment, and Reproducibility

The degradation pipeline and dataset construction are implemented as modular EOTask objects within the `eo-learn` framework, orchestrated through a versioned simulation configuration object rather than hard-coded parameters, so that a given degradation run (baseline-calibrated or extended-sweep) is fully specified by its configuration rather than by code edits. Triplet and LULC datasets are stored in HDF5 with an accompanying manifest recording per-patch provenance and train/validation/test split assignment (fixed random seed, positional split), enabling exact reproduction of any reported split. Model training (TerraMind fine-tuning, DSBN-based distillation, the contrastive-alignment ablation) is implemented in PyTorch. The codebase is organized as a single versioned Python package with submodules separating data simulation, dataset loading, model training tasks, and drift analysis, under standard git version control, so that each experiment reported in Chapter 4 corresponds to a specific, retrievable commit and configuration.

Chapter 4

Results and Discussion

This chapter reports the experiments that evaluate the three research questions of Section 1.3: whether a measurable sensor domain gap exists in the latent space of a geospatial foundation model (RQ1), whether cross-architecture distillation from that model can produce a sensor-invariant compact student (RQ2), and whether doing so is preferable to simply training the compact model on the target sensor directly (RQ3). Following the ablations summarised in Section 4.1, all results below use a single configuration – distillation with domain-specific batch normalization – and the comparison of interest throughout is against a supervised baseline of identical architecture trained without any distillation.

4.1 Experimental Configuration

The teacher is TerraMind [7] at the **base** scale, fine-tuned on the Sentinel-2 branch of the triplet dataset (Section 3.1.1) for the eleven-class WorldCover LULC task [32], and frozen thereafter. The student is a U-Net [33] with 9.85 M parameters, trained from random initialisation; no pre-trained weights are available for this backbone, which is a material constraint returned to in Section ???. Both models consume the seven-band Blue–NIR stack under the per-domain normalization protocol of Section 3.3.1.

Two elements of the configuration deserve to be stated explicitly, because both were arrived at empirically and both materially change the results.

Domain-specific batch normalization. The student carries a separate set of batch-normalization [34] statistics and affine parameters for each sensor, with all convolutional weights shared, following Chang et al. [25]. Beyond its role as a domain-adaptation mechanism, this corrects a train/test inconsistency inherent to the two-branch training scheme: the two sensors are forwarded in separate passes, so during training each pass is normalized by its own domain’s batch statistics, while a single shared set of running averages – a blend of both domains – is what would be used at inference. The encoder would therefore never encounter at test time the normalization under which it was optimised.

Class-imbalanced task loss. The WorldCover class prior on this dataset is severely imbalanced: the three rarest classes (snow/ice, mangroves, moss/lichen) together account for 1.2% of labelled pixels but, under a macro-averaged mIoU, for 27% of the metric. Under unweighted cross-entropy the student never predicted these classes at all, scoring

exactly zero IoU on them in training as well as validation – a failure to learn rather than overfitting. All runs reported here therefore use focal loss [35] with $\gamma = 2$ and inverse-square-root class weighting derived from the measured pixel frequencies, a milder variant of the effective-number scheme of Cui et al. [36].

Contrastive representation-alignment objectives (Section 3.4.3) were implemented and ablated but are disabled in all results below; their weights are zero throughout. That ablation is reported separately and is not the subject of this chapter.

4.2 Sensor Gap in TerraMind Latent Space (RQ1)

The first question is whether the domain gap is visible in the representation itself, rather than only in downstream accuracy. Embeddings were extracted for 800 co-registered test patches from both sensors and compared under three measures: paired cosine similarity between the two views of the same ground, proxy $\mathcal{A}$ -distance (PAD) [22], and a cross-domain probe-transfer test in which a linear classifier fitted on Sentinel-2 embeddings is applied to Φ -sat-2 embeddings of the same patches.

The teacher’s representation does not transfer across the sensor pair. Paired cosine similarity between the two views of the same ground is 0.098 – the two sensors are close to orthogonal in its embedding space – and a linear probe fitted on its Sentinel-2 features retains only 33% of its accuracy when applied to Φ -sat-2 features ($0.754 \rightarrow 0.250$). Figure 4.1 shows this directly: grey lines join the two sensor views of each patch, and in the teacher’s projection they stretch across the entire embedding (median link length 6.15 UMAP [37] units).

Per-layer analysis locates the gap. PAD between the two domains is 1.90–1.98 at every transformer block, including the first, on a scale bounded at 2. A gap that were semantic in origin would be expected to emerge in the deeper, more abstract layers; one present from the first block instead points to low-level radiometry. This is consistent with the spectral analysis of Section 3.3, which found that the two sensors’ digital numbers differ by more than an order of magnitude and that their per-band shapes differ in ways a single gain cannot reconcile. It is worth noting that this ordering is the reverse of the resolution-driven account emphasised in related on-orbit work [11], since Φ -sat-2’s 4.75 m ground sampling distance is finer than Sentinel-2’s.

4.3 Sensor Invariance in the Distilled Student (RQ2)

The distilled student inverts every one of these measurements. Paired cosine similarity rises from 0.098 to 0.948, and probe transfer from 33% to 97% ($0.663 \rightarrow 0.646$): a classifier fitted on the student’s Sentinel-2 features is very nearly as accurate on its Φ -sat-2 features. In Figure 4.1 the paired links collapse to 0.40 UMAP units, a fifteen-fold reduction. The downstream consequence is visible in the right panel of Figure 4.3: the student’s accuracy on the two sensors differs by only 0.032 mIoU on the test split, against a teacher whose representation does not usefully transfer at all.

One measurement in this set behaves counter-intuitively and is worth reporting rather than suppressing, because it illustrates a trap in evaluating this class of method. Raw PAD is maximal for the student (2.000, versus 1.955 for the teacher): a linear probe separates the two domains perfectly. This is an artefact of the mechanism itself. Domain-specific normalization gives each sensor its own affine transform, so a constant per-channel

Same ground, two sensors — grey lines join the paired patches

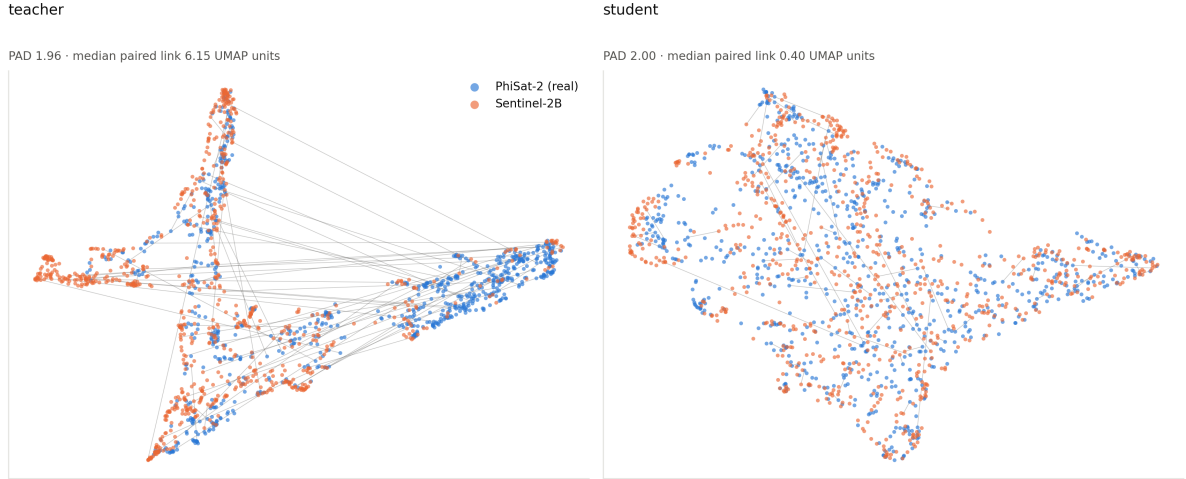


Figure 4.1: UMAP projection of pooled embeddings for 800 co-registered test patches, teacher (left) and distilled student (right), each projected independently. Grey lines join the two sensor views of the same patch. In the teacher’s space the paired views are far apart (median link 6.15 units); in the student’s they are effectively coincident (0.40 units).

offset distinguishes the domains by construction, and a linear probe finds it trivially regardless of how well the semantic content is aligned. Standardising each domain on its own statistics – which removes exactly the first and second moments that a per-domain affine can express – drops the probe to chance for both models. Figure 4.2 shows the underlying quantity: in the raw setting the two domains’ probe scores are separated by an empty margin, and after per-domain standardisation they collapse onto one another.

PAD is therefore not a valid instrument for a model with domain-specific normalization, even though it is scale-free and hence otherwise well suited to comparing representations of different dimensionality. Scale-free is not affine-free. Probe transfer, which asks the functional question directly, is the appropriate measure and is the one reported above.

This result should be stated precisely. With domain-specific normalization the encoder is domain-adaptive rather than domain-invariant: convolutional weights are shared, normalization statistics are not, and the target-sensor branch must be selected at inference. The defensible claim is that after per-domain normalization, structure learned on the source sensor is 97% readable on the target – which is weaker than a single sensor-agnostic encoder, but is the property an on-board deployment actually requires, since the acquiring instrument is always known.

4.4 Distillation Against a Supervised Baseline (RQ3)

Sensor invariance is only useful if it buys accuracy that a simpler method would not. The control is a student of identical architecture trained directly on labelled Φ -sat-2 with no teacher, no second sensor and no distillation, holding constant the split, normalization, augmentation, task loss, optimiser, schedule and early-stopping criterion.

At a budget of 10,000 labelled patches, distillation wins clearly (Table 4.1, Figure 4.3).

What PAD measures: separability along the domain-probe axis

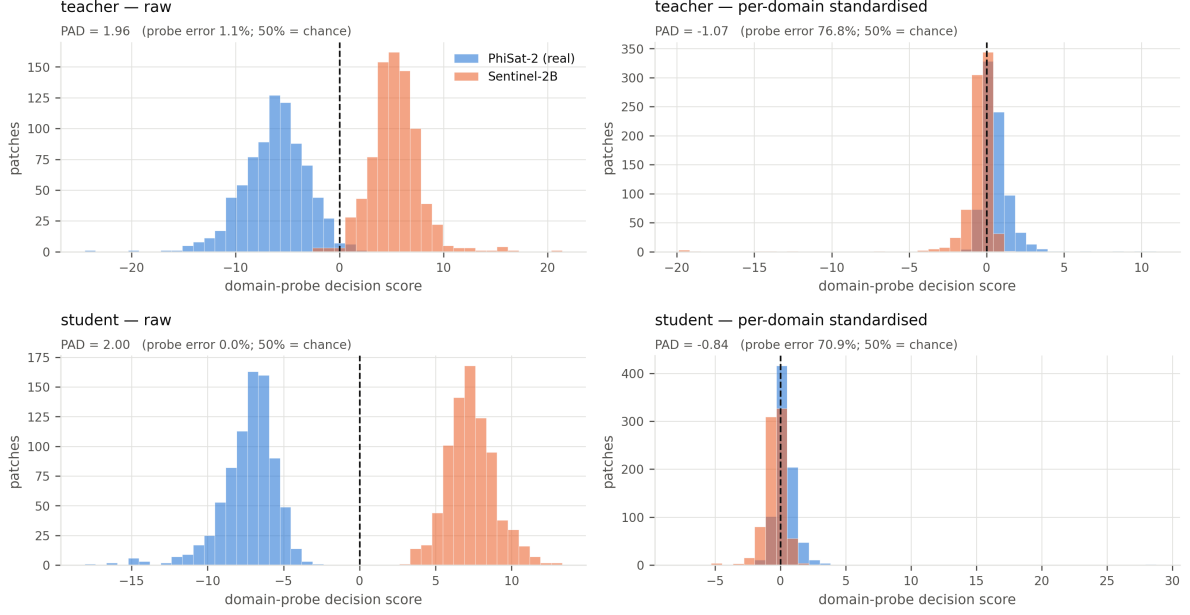


Figure 4.2: Distribution of the domain probe’s out-of-fold decision score, the quantity from which PAD is computed. Separated modes imply a probe that never errs (PAD $\rightarrow$ 2); overlapping modes imply chance (PAD $\rightarrow$ 0). After standardising each domain on its own statistics (right column) the separability vanishes for both models, showing that all of it was per-channel mean and scale rather than representational structure.

Best validation mIoU is 0.4532 against 0.3989, and on matched epochs the distilled student is ahead by 0.0227 mIoU on average, winning 54 of 71 epochs (paired t -test, $p = 1.1 \times 10^{-8}$). The advantage is not present early – the baseline leads for roughly the first twenty epochs – and then widens monotonically, which is why an earlier run truncated at 50 epochs substantially understated it.

Per-class results (Figure 4.4) show the gain is broad rather than concentrated: distillation is ahead on nine of eleven classes, most markedly on mangroves (+0.19), herbaceous wetland (+0.12), grassland (+0.12) and cropland (+0.10). Frequency-weighted IoU moves in the same direction as the macro average (0.5132 against 0.4467), so the result is not an artefact of rare-class weighting.

Two costs qualify this. Distillation requires 7.9 minutes per epoch against the baseline’s 3.3, a factor of 2.4, dominated by the frozen teacher’s forward pass and the second student branch. At a fixed compute budget rather than a fixed epoch count the comparison is much closer, and the baseline may be preferable. Separately, the teacher itself reaches 0.5788 mIoU on Sentinel-2, so the distilled student closes only part of a substantial capacity gap; compression, not merely domain adaptation, remains the binding constraint.

4.5 Advantage Reverse on Small Label Budget

Distillation and unsupervised domain adaptation are usually motivated by label scarcity in the target domain [20], so the expectation is that the advantage above should grow as

Table 4.1: Main results. Validation mIoU is the best epoch; both budgets use the same fixed 1,000-patch validation split. Paired t -tests are over matched epochs.

Labels	Method	Val. mIoU	Test mIoU	Cost (h)
10,000	KD + DSBN	0.4532	0.4254	11.6
10,000	Supervised baseline	0.3989	0.3902	4.4
1,000	KD + DSBN	0.2679	0.2635	2.2
1,000	Supervised baseline	0.2916	0.2707	0.6
TerraMind teacher (Sentinel-2, reference)		—	0.5788	—

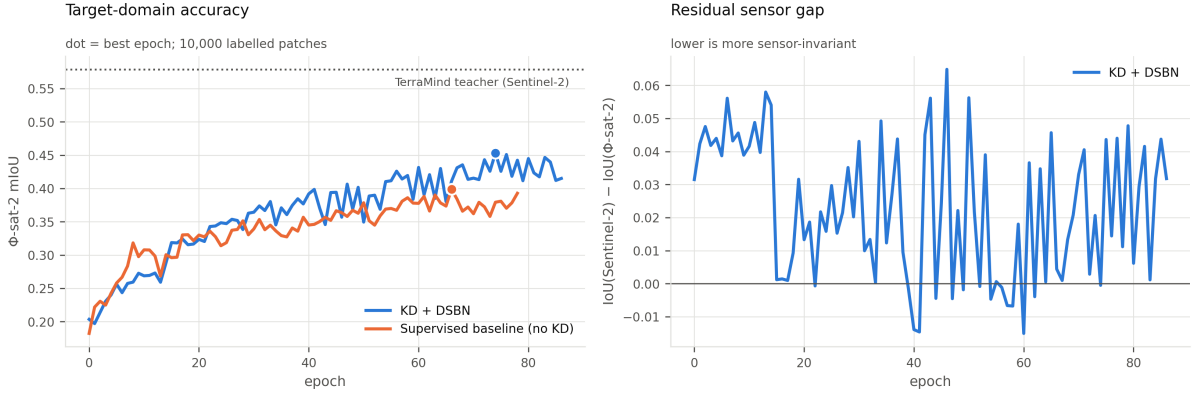


Figure 4.3: Left: target-domain accuracy over training at 10,000 labelled patches; the dotted line marks the teacher’s Sentinel-2 accuracy. Right: the residual accuracy difference between the two sensors for the distilled student, which settles near zero.

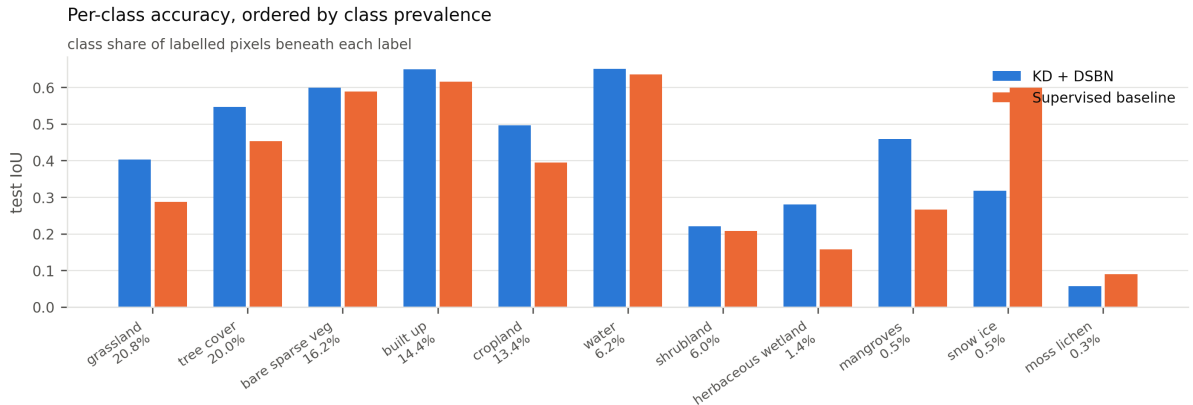


Figure 4.4: Per-class test IoU at 10,000 labelled patches, ordered by class prevalence. Distillation is ahead on nine of eleven classes. Neither model leaves any class unlearned, which unweighted cross-entropy did for four of them.

target labels are removed. It does the opposite.

At 1,000 labelled patches the supervised baseline is ahead by every measure: best validation mIoU 0.2916 against 0.2679, test mIoU 0.2707 against 0.2635, and on matched epochs the distilled student wins only 8 of 58 epochs (paired t -test, $p = 1.5 \times 10^{-14}$). It also does so at roughly a third of the cost, and with the larger budget available to it – the distilled run used its full hundred epochs without triggering early stopping, while the baseline stopped at sixty-seven. Figure 4.5 shows the crossover; both budgets are evaluated on the identical fixed 1,000-patch validation split, so the two points are directly comparable.

Both directions are strongly significant, which makes this a genuine crossover rather than noise at one end. A plausible reading is that the distilled configuration carries substantially more machinery – a second forward pass, doubled normalization parameters, and a distillation term pulling toward a teacher that is itself imperfect at 0.5788 mIoU – and that below some label threshold the cost of that machinery exceeds the value of the teacher’s signal. This is offered as a hypothesis: with two budgets the crossing point is bracketed but not located, and intermediate budgets would be required to characterise it.

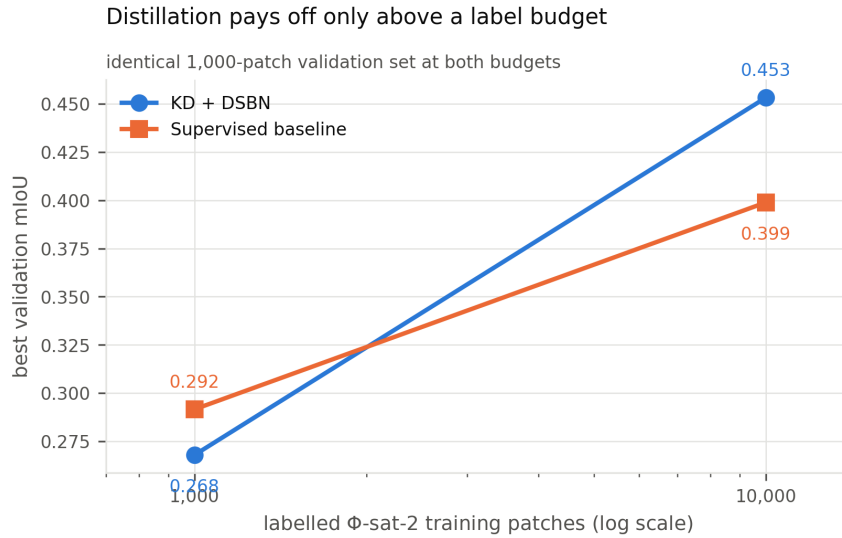


Figure 4.5: Best validation mIoU against labelled training budget. The two methods cross: distillation is worse at 1,000 labelled patches and better at 10,000. Both budgets share the same fixed validation split.

4.6 Discussion

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